Day 098 | Unit 07 | Teacher plan

Understand and Use Percent

Find the Whole from Part and Percent

Standards: 6.RP.A.3c

Learning target: I can find the whole when a part and percent are known.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

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01 / Learn

Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer.

Worked example 1

20% of a number is 16. Find the number and check.

1. Write part = decimal rate × whole.

2. Divide the part by the decimal rate to find the whole.

3. Whole = 16 ÷ 1/5 = 80.

4. Check 20% × 80 = 16.

5. State the result in the form requested: 80.

Answer: 80

Worked example 2

72 is 60% of an unknown quantity w. Select all equations that find w.
A. (60/100)w = 72
B. w = 72 + 60
C. w = 72 ÷ (60/100)
D. w = 72 × (60/100)

1. Write part = decimal rate × whole.

2. The part is percent rate times whole.

3. Divide 72 by 3/5

4. w = 120.

5. Check by multiplication.

Answer: A, C - (60/100)w = 72; w = 72 ÷ (60/100)

Day 098 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

50% of a number is 70. Find the number and check.

02 / Constructed response

10% of a number is 22. Find the number and check.

03 / Constructed response

25% of a number is 40. Find the number and check.

04 / Constructed response

25% of a number is 15. Find the number and check.

05 / Constructed response

15% of a number is 27. Find the number and check.

06 / Constructed response

75% of a number is 75. Find the number and check.

07 / Constructed response

33 students are 15% of a school’s enrollment. Find the enrollment.

08 / Multiple select

12 is 30% of an unknown quantity w. Select all equations that find w.
A. (30/100)w = 12
B. w = 12 ÷ (30/100)
C. w = 12 × (30/100)
D. w = 12 + 30

09 / Multiple select

60 is 60% of an unknown quantity w. Select all equations that find w.
A. w = 60 ÷ (60/100)
B. (60/100)w = 60
C. w = 60 × (60/100)
D. w = 60 + 60

10 / Constructed response

56 students are 40% of a school’s enrollment. Find the enrollment.

Day 098 | Unit 07 | Answer key

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1. 140

Whole = 70 ÷ 1/2 = 140. Check 50% × 140 = 70.

6.RP.A.3c | Constructed response

2. 220

Whole = 22 ÷ 1/10 = 220. Check 10% × 220 = 22.

6.RP.A.3c | Constructed response

3. 160

Whole = 40 ÷ 1/4 = 160. Check 25% × 160 = 40.

6.RP.A.3c | Constructed response

4. 60

Whole = 15 ÷ 1/4 = 60. Check 25% × 60 = 15.

6.RP.A.3c | Constructed response

5. 180

Whole = 27 ÷ 3/20 = 180. Check 15% × 180 = 27.

6.RP.A.3c | Constructed response

6. 100

Whole = 75 ÷ 3/4 = 100. Check 75% × 100 = 75.

6.RP.A.3c | Constructed response

7. 220

Whole = 33 ÷ 3/20 = 220. Check 15% × 220 = 33.

6.RP.A.3c | Constructed response

8. A, B - (30/100)w = 12; w = 12 ÷ (30/100)

The part is percent rate times whole. Divide 12 by 3/10; w = 40. Check by multiplication.

6.RP.A.3c | Multiple select

9. A, B - w = 60 ÷ (60/100); (60/100)w = 60

The part is percent rate times whole. Divide 60 by 3/5; w = 100. Check by multiplication.

6.RP.A.3c | Multiple select

10. 140

Whole = 56 ÷ 2/5 = 140. Check 40% × 140 = 56.

6.RP.A.3c | Constructed response